

2(a)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
2(b)(i)	evidence of two temperature conversions between °C and K	B1
	two calculations shown, one for each state e.g. $\frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{(273 + 27)} = 0.198 \text{ and } \frac{6.70 \times 10^6 \times 30 \times 10^{-6}}{(273 + 742)} = 0.198$	A1
2(b)(ii)	work is done on the gas	M1
	internal energy increases (so temperature increases)	A1

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Question	Answer	Marks
2(b)(iii)	$pV = NkT$ e.g. $N = \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$ $= 1.435 \times 10^{22}$ $\Delta E_k = (3/2) k \Delta T N$	C1
	$= (3/2) \times 1.38 \times 10^{23} \times (742 - 27) \times \frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$	C1
	$= 212 \text{ J}$	A1
2(c)	$E = mc\Delta\theta$ and $E = mL$	C1
	$\Delta\theta = (27 + 196)$ or 223	C1
	$E = 0.0240 \times 1.04 \times (27 + 196) + 0.0240 \times 199$ $= 10.3 \text{ kJ}$	A1